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Physics-Based Event Attribution From Sparse PMUs With Recovery Memory

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ABSTRACT An electrical network can continue oscillating after a load is restored or a line is reclosed. A later intervention can then resemble the recovery of an earlier one. We address this ambiguity by comparing physical histories from eight sparse PMUs. DAE-derived switched predictors use voltage, oriented terminal current, frequency and ROCOF, start from estimated past states, and preserve recovery through restoration. A ten-second state history supports screened onset proposals; conditional evidence integrates load severity and updates fixed network hypotheses. Active-event and originating-cause masses keep ongoing interventions separate from their remaining response. Completed development replays with a common publication clock give 51/54 timely family/source decisions with 3 extra alarms, versus 37/54 and 37 for a simpler state-retaining physical procedure. Independent confirmation is in progress. The contribution is a specified recovery-aware attribution mechanism and an evaluation separating detection, timely location and restoration. The frozen physical startup contract rejects the retrospective RAW reference; a separate learned source hybrid corrects only one additional physical window. The supported scope is a known load/line catalog; probability calibration, all-family diagnosis and real-time deployment remain unestablished.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.[1]

Physical restoration is not electrical recovery. A detector that returns immediately to a flat reference can issue another alarm for the remaining oscillation. Suppressing alarms throughout recovery can instead hide a real intervention. The operating point adds another difficulty: correcting a voltage or current offset need not correct how the network responds to a later input.

Existing work provides strong foundations. DAE state estimators infer unmeasured electrical states [2]; physical outage detectors compare network hypotheses sequentially [3], [4]; and simulated fault trajectories support diagnosis from sparse PMUs [5]. Physical models, transient histories and unknown-onset searches are thus established ingredients. A small state-reconstruction error, however, does not by itself identify the intervention that produced the observations.

We compare physical histories rather than isolated signal features. A history specifies an intervention, source, onset, severity and restoration. DAE-derived predictors start from estimated past states and preserve the state left by previous interventions. Conditional evidence integrates load severity, accounts for instrumental memory, and updates fixed network-model hypotheses. Separate active and cause masses keep a restored intervention available as an explanation of the continuing response.

The paper makes three contributions within this restricted problem:

1) An executable attribution mechanism that makes new load/line interventions compete with retained recovery histories. We specify its physical predictors, evidence updates and decision rules.

[1]Drafting/revision assistance: OpenAI Codex [1].

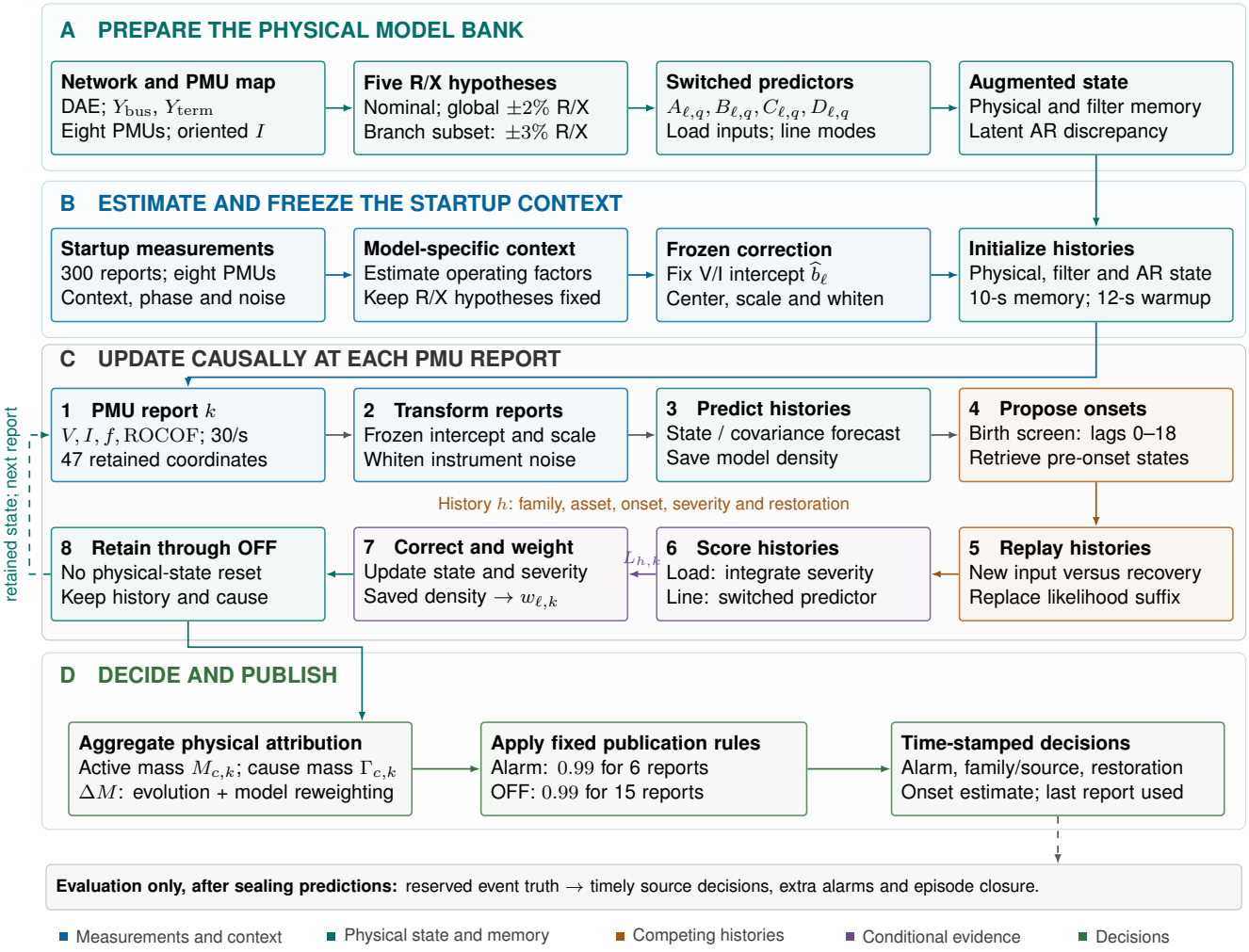


FIGURE 1. Implemented physical-history attribution with its original publication policy, organized into model preparation (A), startup context (B), per-report inference (C), and publication (D). The bank predicts before state correction and saves model evidence before proposal selection. Candidate replay replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Reserved event truth is used only after predictions are sealed. The common-clock controls replace block D as specified in Section VII.

2) A comparison that separates detection, correct source attribution before restoration, extra alarms and episode closure. Nominal and uncertain network models are evaluated alongside two fixed ExtraTrees adapters; Chen's published detector is compared on eligible persistent line openings.
3) Adversarial tests of scope: a state-retaining physical competitor, a classifier fitted under a grouped SIM split, and a retrospective integrity/source replay. These tests retain failures and distinguish simulation utility from transfer to the reference record.

A further physical control asks whether recovery increases source separation. It distinguishes retaining a consistent prediction from obtaining additional information about the source; it is not a comparison of causal detectors.

On the common development cohort and publication clock, the five-model bank obtains 51/54 timely family/source decisions with 3 extras, versus 37/54 and 37 for the simpler physical control. This compares complete procedures; it does not isolate a causal effect of history retention. The prespecified confirmatory comparison is still running without method changes. The claim is scoped to the declared load/line catalog. These simulations do not establish an all-family hackathon score, calibrated event probabilities or real-time deployment.

II. RELATED WORK

A. STATE ESTIMATION AND PHYSICAL SOURCE INFERENCE

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [2], [6], [7]. Functional observability asks whether a requested state quantity can be recovered [8], [9]. Source inference asks which intervention can explain the trajectory. Descriptor-model diagnosability and nuisance-profiled tests formalize this distinction [10], [11].

Physical hypotheses support outage inference from phasor

angles [12], Bayesian inference with uncertain states [13], and sequential detection under stochastic operation [3], [4]. Anguluri *et al.* recover sparse forcing locations, inputs and an unknown initial state using group LASSO for linear dynamics [14]. These are precedents for source recovery; group LASSO is not an executed comparator in our frozen campaign. Sensor placement also affects source distinguishability [15]–[17].[2]

B. TRAJECTORY-BASED AND SEQUENTIAL DIAGNOSIS

Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window and return candidate lines [5]. Their evaluated target is short-circuit identification before clearing; load changes are an adverse-event test. Our target follows load changes and line openings through restoration and successive interventions. Physical trajectory libraries, remote PMUs and retrospective onset scanning are shared ingredients. Their procedure was not run on our data, so no numerical superiority is inferred.

The Adaptive Matrix CuSum derives sequential identification error/delay results under specified observation laws [18]. Multiple-model filtering already propagates conditional state/mode hypotheses [19]. The multiple-model multiple-hypothesis filter maintains mode-sequence histories and adapts their number through pruning and merging [20]. Retaining histories or adapting bank size is therefore not an original principle of our method. Our pruned history bank uses these ideas, but its approximate covariances, proposal screen and fitted startup context do not inherit their general guarantees. The complete procedure evaluated here makes retained recovery compete with later interventions and reports timely attribution and alarm burden. These comparisons do not isolate recovery memory from the other admission and publication rules.

C. LEARNED DIAGNOSIS AND PHYSICAL UNCERTAINTY

Learned and knowledge-driven representations identify PMU disturbances and assets [21]–[23]. We compare two fixed ExtraTrees adapters from the SGSMA project. Their performance tests transfer of those implementations, not the limits of all supervised learning. A known physical catalog and source-excluded supervised training also impose different information constraints.

Physical stochastic models support latent-state estimation [24], [25], while joint DAE parameter inference addresses uncertain generators and networks [26]. Our stream method uses a finite bank of constant network hypotheses. Trajectory sensitivities and Gaussian evidence have also been combined for Bayesian parameter inference in power-system DAEs [27]. Their use alone does not distinguish an event attribution algorithm. The separate signature test examines a local event–context derivative; it does not identify every plant parameter or replace the stream method with an unrestricted joint estimator.

[2]Drafting/revision assistance: OpenAI Codex [1].

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.[3]

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_q(x_k, z_k; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The measurement map includes gains, phase references, orientation and the causal frequency/ROCOF estimator. A machine's rotor speed is not substituted for bus frequency.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

[3]Drafting/revision assistance: OpenAI Codex [1].

TABLE 1. Closest inference contracts. Only Chen and the fixed ExtraTrees adapters are executed comparators in the confirmatory stream campaign. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

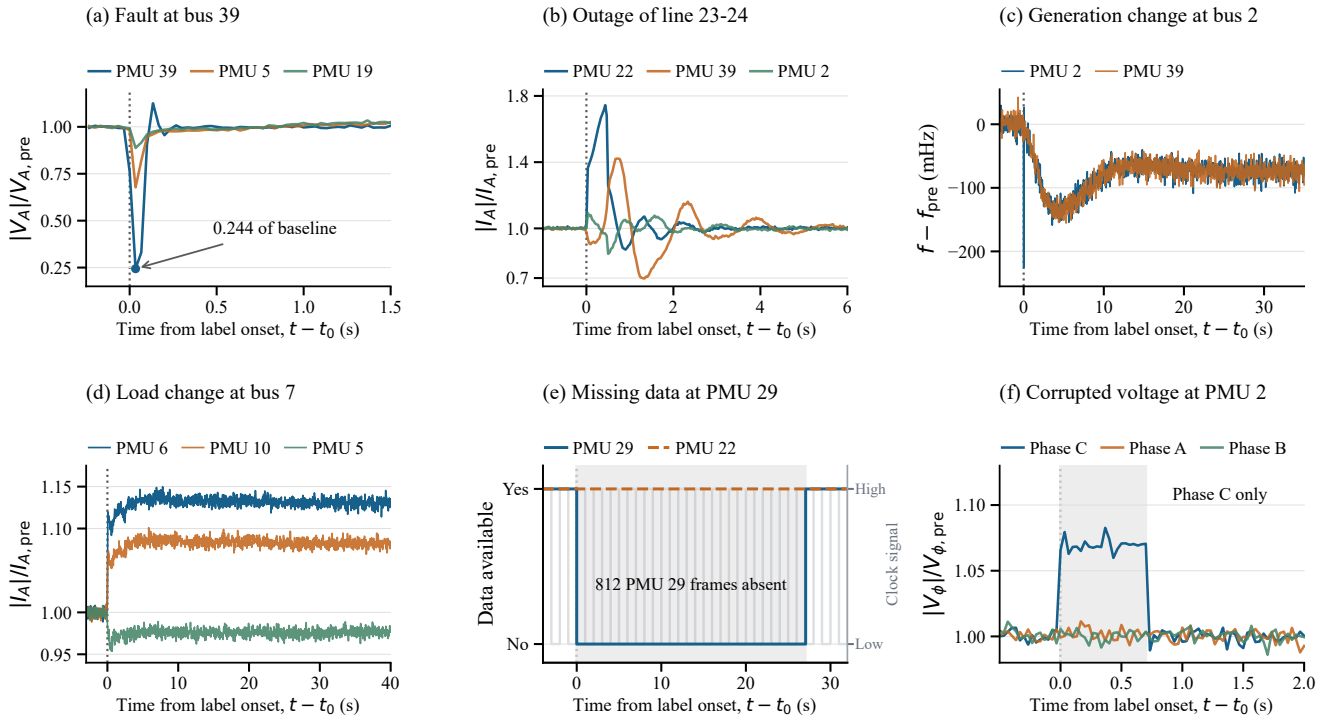


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine, archived as R11, uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.[4]

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insuffi-

[4]Drafting/revision assistance: OpenAI Codex [1].

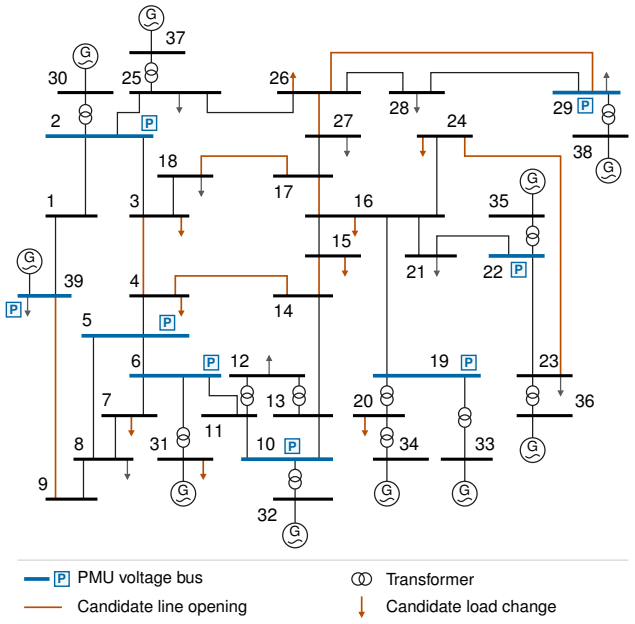


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

TABLE 2. **Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.**

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside C01
Unknown	Unresolved diagnosis	Not a new physical family

cient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u), \\ \delta\dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \overline{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

The supplement derives the closed form by completing the square and works through a numerical example. It also specifies the conditional state representation $m_h + d_h a$ and its update. The local amplitude integral is exact for the stated conditional Gaussian model. The complete bank remains approximate: regime covariances are shared from nominal-operator Riccati equations, context estimates enter by substitution, and histories are screened and pruned. At episode closure the augmented state mixture mean is retained and omitted between-history spread is recorded. A fully Bayesian posterior over states, context and all histories is not claimed.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{11}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t, \kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{12}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a trajectory, not a zero residual; it is not a new propagation theorem.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{13}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\begin{aligned} \Delta M_{c,k} = &\sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ &+ \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \end{aligned} \tag{14}$$

The terms distinguish within-model evolution from model reweighting. This accounting identity aids interpretation, without constituting a new tensor factorization. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

The supplement records the frozen birth/restoration hazards, severity prior and integration intervals, restoration-child rules and beam limits, together with the execution sequence and illustrative calculations. These details preserve the archived configuration; the compact algorithm below states the report order and information boundary.

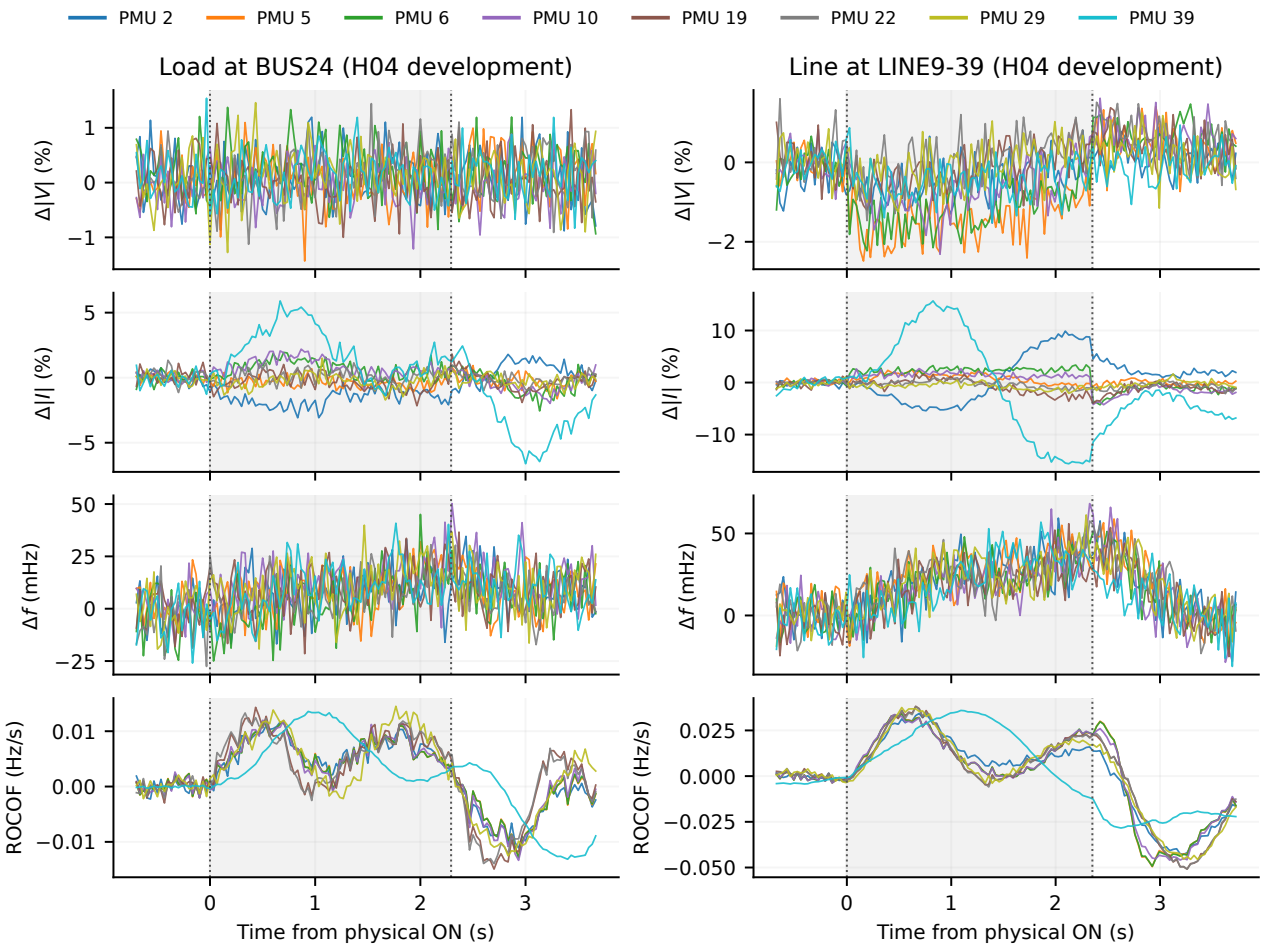


FIGURE 4. **Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

G. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the frozen short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.[5]

The simulator is the IEEE-39 Julia/PowerDynamics research candidate F01. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

H03 and H04 each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

R11 and its nominal version are frozen co-references. The two archived ExtraTrees artifacts (the R09 references) and their preprocessing/publication adapters are also frozen; neither is retrained on H04 or C01. They receive the same 30-fps observations. The separately fitted S01 SHORT classifier uses the grouped SIM contract described below and is then frozen for H04 and RAW. It does not replace the archived comparators in C01. These are comparisons of specified implementations, not a hackathon leaderboard rerun.

The published physical comparator is Chen *et al.*'s persistent-line quickest-change detector [3], which considers single- and double-line outages. Our declared F01 adapter restricts its fixed nominal catalog to single-line hypotheses. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. The adapter uses nonoverlapping angle increments at 15 Hz and a startup-only scale/drift estimate. Its white-injection/DC assumptions do not describe all F01 instrumental effects. We preserve both matched-law implementation checks and F01 mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign, C01, therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

C01 specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent evaluation is in progress. The following section reports completed development results only.

VI. CONTINUOUS-STREAM RESULTS

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

This section uses each frozen method's original publication policy. The common-clock physical control in Section VII republishes the same history-bank outputs under a different declared rule; its counts are a separate endpoint. The ExtraTrees rows here are the archived R09 references, distinct from the newly fitted S01 classifier.[6]

In H04, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 4). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the frozen adapters; it neither rules out supervised load detection nor reproduces the historical multiclass hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 6 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later

[5]Drafting/revision assistance: OpenAI Codex [1].

[6]Drafting/revision assistance: OpenAI Codex [1].

TABLE 3. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Signature component test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Recovery geometry control	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; not R11 performance
H04 procedure controls	Same development parents and observations as above	Single-state GLR, history banks and newly fitted ET	Common event metrics; distinct score/publication semantics
Grouped SIM learning	31/8/17 fit/validation/reserved groups; 90/16/70 records	Frozen classifier selection and reserved performance	Previously open campaigns; no unseen-source claim
RAW reference replay	One retrospective record; 180 fixed windows, 24 physical-positive	Physical admission, integrity timing and source composition	Metadata-based identities; no exact actuation log or independent field claim

TABLE 4. H04 with the original publication policies. ExtraTrees rows are archived R09 comparators. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
Archived ET ref.	0/27	27/27	3	0
Archived ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and (14) separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative experiment leaves R11 and nominal frozen for C01.

VII. ADVERSARIAL CONTROLS AND THE TRANSFER BOUNDARY

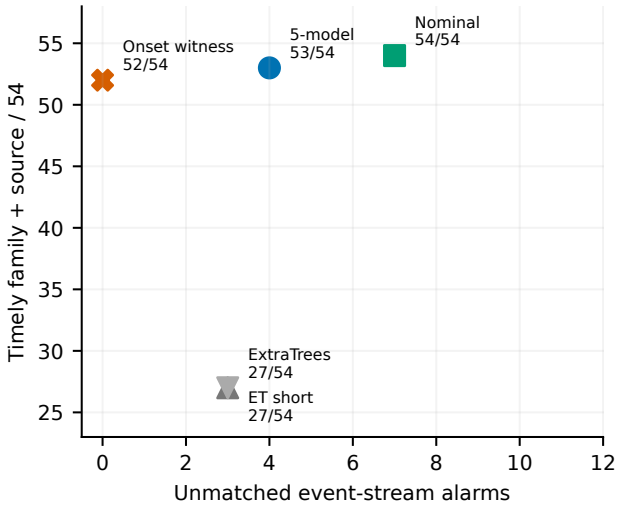


FIGURE 5. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

A. A SIMPLER PHYSICAL COMPETITOR

A stronger simplicity check retains one committed physical and instrumental state, rather than resetting to equilibrium after each event. Short GLR alternatives test the same admitted load and line operators over at most 19 reports. Load ON severity is profiled within the original bounds; an accepted OFF changes the operator while retaining the inferred state.

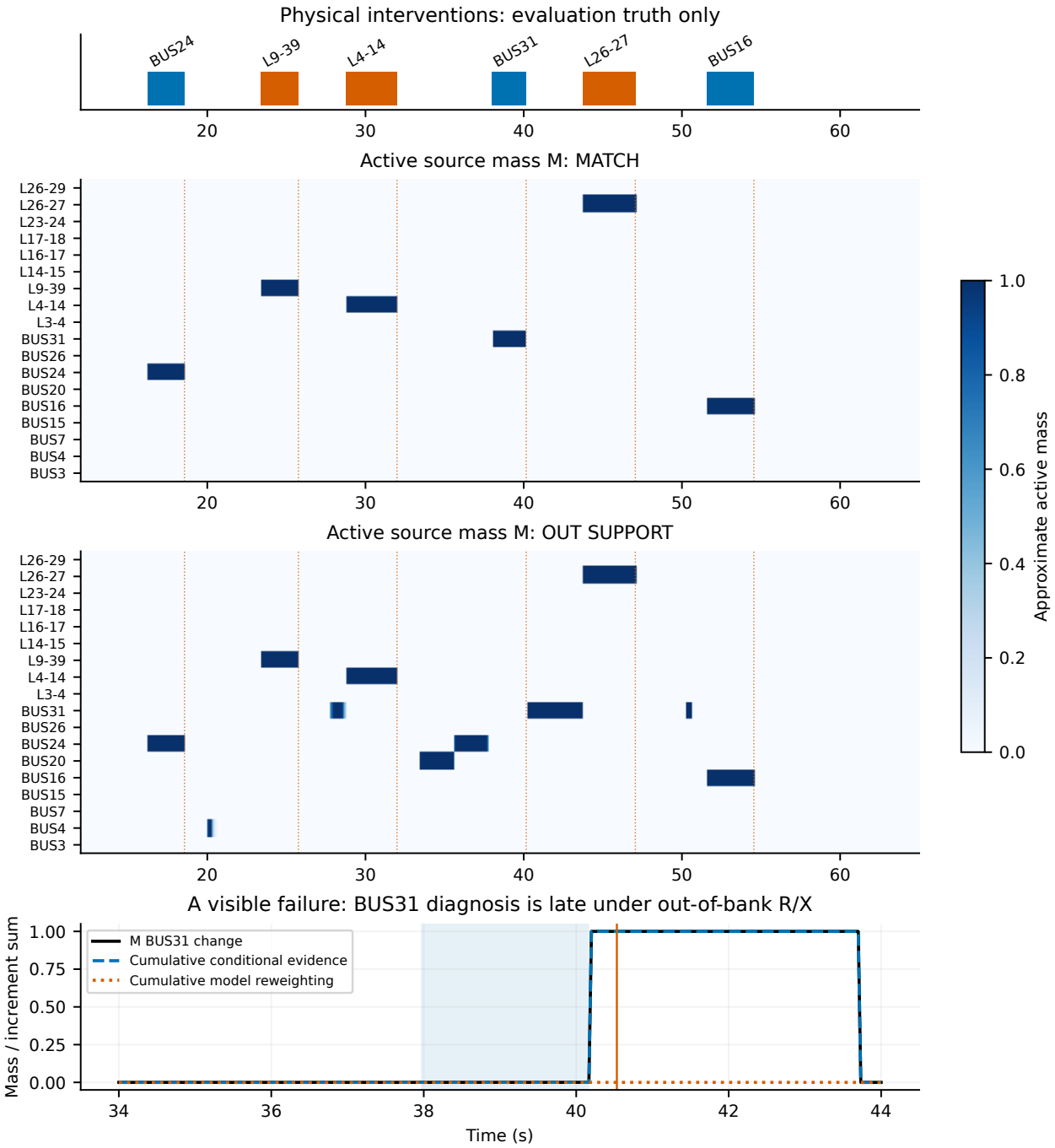


FIGURE 6. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in development parent H002. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

TABLE 5. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

Alternative histories then expire. This is a complete simpler procedure, not a deliberately memoryless reset baseline.

The single-state procedure and nominal history bank share fitted nominal operators; the five-model bank additionally mixes the declared resistance and reactance alternatives. All use a six-report source-stability clock. The two archived history-bank outputs are republished with that clock. The GLR includes a fixed penalty for its maximum search count; neither this penalty nor a numerically shared threshold makes its scores posterior probabilities or equalizes false-alarm rates. Table 5 aggregates all twelve records. The five-model bank obtains 51/54 timely decisions with 3 extras, versus 37/54 and 37 for the single-state procedure. This favors the full procedure in this development contrast, but does not isolate history averaging from severity integration, search or regime selection. These are three event calendars with repeated conditions. The zero alarms over 204 normal seconds are too little exposure to establish a low operational alarm rate. The changed counts relative to the original publication policy also show that the decision clock is consequential. The corresponding orphan-restoration counts are 32, 5 and 3 for the single-state, nominal and five-model procedures; zero premature matched closures does not imply error-free restoration. The original publisher gave 53/54 with 4 extras for the mixture and 54/54 with 7 for nominal. The common publisher gives 51/54 with 3 and 44/54 with 6, respectively. These are different decision policies on the same H04 outputs, not conflicting estimates of one endpoint.

B. WHAT GROUPED SIM TRAINING ESTABLISHES

We also fit two ExtraTrees variants using dependency groups, rather than randomly splitting correlated reports. The approximately 70/30 split is subdivided into 31 fitting, 8 validation and 17 reserved groups. SHORT is selected on validation and frozen before the reserved evaluation. Table 6 retains its marked load/line asymmetry. Its 302/576 timely reserved decisions and 3 extras do not establish adequate general transfer. The corpus was already open and all physical sources remain represented in development; this is neither a blind test nor an unseen-source split. Only one reserved normal dependency group underlies the 3696 normal seconds with 0 alarms.

Replaying the selected classifier on the common H04 cohort gives 27/54 timely diagnoses and 3 extras: 0/27 loads and 27/27 lines. The physical bank's benefit therefore lies chiefly in loads in this contrast; the classifier resolves all tested lines. Its frozen publisher uses six identical argmax labels, without the physical procedures' score threshold. This is a common-cohort, common-metric comparison of complete procedures, with different publishers. H04's normal parent also appears in the reserved SIM fraction and is not additional independent normal evidence.[7]

TABLE 6. Grouped SIM development. SHORT is selected only on validation; the reserved 30% is evaluated once. These cohorts are distinct from H04.

Variant and partition	Load	Line	Extras
Baseline, validation	5/96	87/96	0
SHORT, validation	16/96	92/96	0
SHORT, reserved	28/288	274/288	3

C. MEASUREMENT INTEGRITY AND RAW0001

The frozen physical filter does not admit RAW0001: the initial prefix violates its ROCOF noise-scale and temporal-stationarity support. The following ZIP and segmented-ML replays therefore examine complementary integrity and source interfaces; they do not validate R11 transfer. The submitted multi-family artifact is replayed on all 180 fixed windows of RAW0001. Its predictions are emitted at window completion. A separate availability monitor reports missing PMUs from received flags and finite-value masks; it keeps missing-PMU identity, corrupt-PMU identity and physical source distinct. Such diagnoses do not establish malicious intent. Complete-window corruption diagnoses describe their observed interval and are not broadcast backwards as causal labels.

Against the current annotation support, the submitted artifact detects an abnormality in 21/39 positive windows, with 1 false-positive window among 141 normal windows; all 180 windows produce outputs. Correct physical-family coverage is 4/24. The availability monitor identifies all 9200 missing reports and their PMU, whereas corruption localization is correct in 1/2 positive windows. A separate spurious corruption window is retained. These denominators describe one retrospective record, not independent field trials. Both implementations identify all 4 missing intervals and their PMU; the immediate interface reduces mean first-notice delay from 12.13 to 0 seconds on the received-report clock. This limited publication benefit does not measure wall-clock latency or physical inference. Its event-code annotations do not certify individual physical actuation boundaries, so actuation delay and restoration are not scored.

The archived source spreadsheet permits a separate metadata-based source comparison. A frozen hybrid uses the SIM-trained SHORT classifier only to replace the submitted artifact's load/line source when a first same-family diagnosis has already been emitted within that window. Predicted mixed physical/missing cases are eligible under the same rule. All classes, integrity diagnoses and window emission times remain fixed. Missing input splits the ML stream into complete segments with explicit unavailable startup; their

[7]Drafting/revision assistance: OpenAI Codex [1].

TABLE 7. Retrospective RAW family-and-source correctness on every physical-positive window, using the archived source metadata and current event-code support. Multiple windows from a sustained annotation are not independent actions.

Family	Submitted ZIP	Source hybrid
Fault	1/1	1/1
Line	0/1	1/1
Generation	0/11	0/11
Load	0/11	0/11
Total	1/24	2/24

prefixes are not verified normal. This is a segmented ML transfer test, not a successful execution of the physical history bank. Joint correctness changes from 1/24 to 2/24: 1 paired gain and 0 losses, with 179/180 total windows retaining the ZIP source. The gain corrects the line location; load and generation remain unresolved. A single improved window in one previously used record does not establish reliable all-family transfer.

The implemented interface combines these physical and integrity outputs in one chronological stream. Exact projection checks recover all 180 component windows and the availability state across 161379 reports. This verifies composition without adding a performance gain; completed-window evidence cannot reopen an already restored missing-PMU episode.

The calendar recovered for the archived hackathon table does not match the actual RAW annotations; its score is not a reproduced comparator. Relabeling windows or relaxing one admission gate would not validate the physical/instrument model. The archive preserves these failures and the current-calendar evaluation. Superiority over the historical all-family system remains unestablished.

VIII. REFERENCE AND RESPONSE ARE DIFFERENT CORRECTIONS

A separate controlled experiment tests a component of the local predictor: whether correcting the PMU reference is sufficient when operation changes. It compares a fixed signature, that signature with the measured prefix offset, and an operating-conditioned signature with the same offset. All methods receive the load family and ON/OFF times. Conditioning reduces median relative severity error, evaluated at the generating source, from 1.2% to 0.329%, while matching reference-only source accuracy. Thus better response fidelity does not automatically improve source decisions, and these oracle-context results are not another unknown-onset detector score. The supplementary material gives the mixed derivative, full scenario design, paired results, nominal-condition counterexample and signal plots. This component experiment motivates conditional signatures without enlarging the causal stream claim.[8]

IX. INTERPRETATION AND LIMITS

[8]Drafting/revision assistance: OpenAI Codex [1].

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{15}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{16}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [10], [11].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. RECOVERY IS NOT A UNIVERSAL SOURCE-INFORMATION GAIN

A separate full-F01 control supplies the prior history, later onset, family and candidate source pair. Profiled source sep-

aration is essentially unchanged under the known history and decreases when the tested prior-severity uncertainty is added; the binary source comparison gains no mean accuracy. The full scenarios, figure and numerical results are retained in the supplementary recovery control. This negative result supports retaining recovery for prediction consistency, without claiming that it universally adds source information or proves an R11 performance gain.

C. OPERATIONAL MEANING AND SCOPE

The physical distinction underlying the stream method is simple: restoration changes an input or topology, but does not erase the machine state. Retaining that state lets recovery compete with a new intervention. The development comparison also shows why alarm count and timely location must be reported together. Stricter admission can suppress extras while rejecting a real load. The existing admission ablation does not isolate the contribution of every recovery-memory operation.

The signature experiment addresses a different source of error. Its corrected variants receive the same pre-event measurements and offset. Any difference between them therefore comes from the event–context term and its estimated context, rather than a privileged baseline or longer window. This isolates the tested response approximation, but not an improvement of the unchanged causal detector. Known timing, one scalar context and four load sources make it a controlled component test.

Our construction builds on trajectory-based diagnosis, sequential likelihoods and multiple-model filtering. Their general principles are not new contributions here. The methodological claim concerns the specified competition among intervention and recovery histories; the experiments measure its operational behavior. Numerical superiority is claimed only where a common-data comparison supports it, not over Warner or other procedures that were not executed.[9]

The continuous physical catalog contains nine load buses and nine lines and requires an intervention-free startup. Its operation through missing PMUs, unknown families, source-excluded causal transfer and field conditions is unvalidated. The separate retrospective integrity interface in Section VII does not extend that physical catalog. F01 retains a provisional BUS5 current terminal and excludes BUS39 current angle. Agreement with this research model does not establish the producer's CT placement or exact reproduction of RAW0001. Local predictors, history pruning, covariance approximations and fitted startup parameters also limit probability claims.

F01's noise port applies temporal and cross-PMU structure within each signal family. Its archived cross-family factor covariance is not applied, and frequency/ROCOF noise is added after phase estimation. The inference comparison therefore concerns this declared simulator, not the complete dependence structure of a physical PMU installation.

[9]Drafting/revision assistance: OpenAI Codex [1].

The two ExtraTrees adapters are particular frozen implementations. Their load-transfer failure does not rule out learned load diagnosis or establish a better official hackathon score. Chen's eligible test concerns persistent line openings; its F01 adapter has DC/white-injection assumptions that do not capture all instrumental effects. Its mismatches and censored decisions remain part of the comparison.

Joint context uncertainty, unrestricted simultaneous sources and calibrated posterior probabilities remain unvalidated. Such extensions must preserve source-discriminating response directions and account for dependence between estimation error and measurement noise.

Finally, processing reports causally is not proof of real-time throughput. Observation time, computation and queue delay must be measured separately. The present implementations remain research inference services rather than validated protection or control systems.

X. CONCLUSION

We formulated sparse-PMU event attribution through competing physical histories. Load/line predictors start from estimated past states and retain the cause of the response after restoration. Active mass M and its increment ΔM expose the changing explanation rather than replacing the physical dynamics.[10]

Under the common development publication clock, the mixture obtains 51/54 timely family/source decisions with 3 extras, versus 37/54 and 37 for the simpler state-retaining physical control. Independent confirmation remains in progress.

The frozen physical procedure rejects the RAW startup contract. The separate integrity and learned-source interfaces provide only limited retrospective improvements, preserving the transfer boundary.

The result is a specified physical attribution procedure and a testable separation of recovery, reference error and source discrimination. Its scope remains the declared simulation contracts. It does not establish all-family diagnosis, universal source recovery, calibrated probabilities or real-time deployment.

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[11] Biographical drafting assistance: OpenAI Codex [1].